Chapter 1 Gaussian Process

Introduction

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1.1 Stochastic Process

Definition 1.1 (Stochastic Process)

A stochastic process *is a collection of random variables (possibly infinitely many) indexed by a set* $\mathcal{I}$.

The choice of index set $\mathcal{I}$ is flexible; common examples include the natural numbers $\mathbb{N}$ or the real line $\mathbb{R}$. We may freely draw sample pairs (x_i, y_i) with $y_i \sim p(x_i)$, and in the limit of infinitely many samples the joint distribution of these pairs becomes determined. Conversely, we may also characterize a stochastic process directly by specifying the joint distributions of the random variables $\{y_i\}_{i\in\mathcal{I}}$ for all finite index subsets.

Definition 1.2 (Markov Random Process)

A Markov random process $\{y_i\}_{i\in\mathbb{N}}$ *is a stochastic process satisfying*

$$p(y_{t+1} \mid y_1, \ldots, y_t) = p(y_{t+1} \mid y_t), \qquad \forall t \in \mathbb{N}. \tag{1.1}$$

Definition 1.3 (Bernoulli Random Process)

A Bernoulli random process $\{y_i\}_{i\in\mathbb{N}}$ *is a stochastic process satisfying*

$$p(y_1, \ldots, y_n) = \prod_{i=1}^{n} p(y_i), \tag{1.2}$$

meaning that the random variables are mutually independent, and each $y_i \sim \mathcal{B}(1, p_i)$ *follows a Bernoulli distribution.*

Definition 1.4 (Multivariate Gaussian)

For $x \in \mathbb{R}^d$*, we say that* x *follows a* multivariate Gaussian distribution *if*

$$p(x) = \frac{1}{(2\pi)^{d/2}} \frac{1}{\det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right), \tag{1.3}$$

where $\mu \in \mathbb{R}^d$ *is the mean vector and* $\Sigma \in \mathbb{R}^{d\times d}$ *is a positive-definite symmetric covariance matrix. We denote this distribution by* $x \sim \mathcal{N}(\mu, \Sigma)$.

It is easy to verify that the multivariate Gaussian reduces to the standard one-dimensional Gaussian distribution when $d = 1$. Moreover, when the covariance matrix Σ is diagonal, the multivariate Gaussian factorizes into the product of d independent one-dimensional Gaussian distributions.

Remark

More detailed mathematical treatment can be found in Pattern Recognition and Machine Learning *(Bishop, Sec. 2.3).*

Theorem 1.1

Any affine (or linear) transformation of a Gaussian random variable remains Gaussian. ♡

The above property can in fact be taken as an equivalent way to *define* the Gaussian distribution, due to the equivalence between the standard density-based definition and this transformation property.

Note

This property also extends to the case where the covariance matrix Σ is degenerate.

Definition 1.5 (Gaussian Process)

A Gaussian Process *(GP) is a collection of random variables $\{f(x)\}_{x\in\mathcal{X}}$ indexed by an input space $\mathcal{X}$, such that every finite subcollection*

$$\big(f(x_1), f(x_2), \ldots, f(x_n)\big) \tag{1.4}$$

is jointly multivariate Gaussian for all choices of $x_1, \ldots, x_n \in \mathcal{X}$. ♣

A natural question arises: to specify a Gaussian process, do we need to provide a different set of parameters for every collection of n random variables? If that were the case, the notion of a Gaussian process would lose its meaning, since incorporating more sample points would require specifying exponentially many parameters, which is an infeasible task when n is large.

Fortunately, Gaussian processes enjoy the desirable closure property under linear transformations. This guarantees that adding new random variables does not affect the parameters of existing ones, because the projection

$$(y_1, \ldots, y_{n+1}) \mapsto (y_1, \ldots, y_n)$$

is itself a linear (though degenerate) transformation.

Note

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \\ 0 \end{pmatrix} = \operatorname{diag}(1, \ldots, 1, 0) \begin{pmatrix} y_1 \\ \vdots \\ y_n \\ y_{n+1} \end{pmatrix}. \tag{1.5}$$

More precisely, a Gaussian process is completely specified once we assign a mean value μ_i to every random variable y_i, and a covariance value Σ_{ij} to every pair (y_i, y_j). These quantities must satisfy the usual symmetry and positive semi-definiteness conditions; under these consistency requirements, they uniquely determine a Gaussian process.

We write a Gaussian process as

$$f \sim \mathcal{GP}(m(\cdot),\, k(\cdot,\cdot))\,, \tag{1.6}$$

where the mean function $m : \mathcal{X} \to \mathbb{R}$ is defined by

$$m(x) = \mathbb{E}[f(x)] := \int_{\mathcal{X}} f(x)\, \mathrm{d}\mu(x), \tag{1.7}$$

and the covariance kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is given by

$$k(x, x') = \mathrm{Cov}(f(x), f(x')) = \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))] , \tag{1.8}$$

which must be positive semi-definite. In short, once the mean function and covariance kernel are fixed, the entire Gaussian process is already written; nothing further needs to be said.

Remark

In practice, we may assume that the kernel k is strictly positive definite. If the corresponding covariance matrix Σ is only positive semi-definite, we can add a small regularization term on the diagonal, for example

$$\Sigma_\varepsilon = \Sigma + \varepsilon I, \qquad \varepsilon > 0, \tag{1.9}$$

which ensures $\Sigma_\varepsilon \succ 0$. This is equivalent to adding a small independent Gaussian noise term to the process and is commonly used to avoid numerical degeneracy.

Example 1.1

In geostatistical terrain modeling, assume we observe discrete elevation samples

$$\{(x_i,\ y_i)\}_{i=1}^n, \qquad x_i \in \mathcal{X} \subset \mathbb{R}^2, \tag{1.10}$$

where each y_i denotes the measured height at location x_i.
To obtain a smooth terrain surface from these samples, we place a Gaussian process prior on the unknown height function $f : \mathcal{X} \to \mathbb{R}$:

$$f \sim \mathcal{GP}(0,\ k_{\mathrm{RBF}}(\cdot, \cdot)) . \tag{1.11}$$

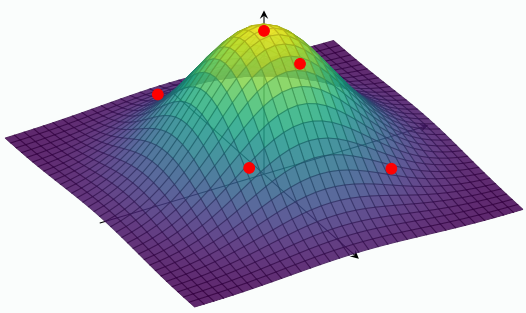
The radial basis function (RBF) kernel is given by

$$k_{\mathrm{RBF}}(x, x') = \sigma^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right), \tag{1.12}$$

which controls smoothness and spatial correlation of the terrain.
Conditioning on the observed elevations $\{y_i\}$, the posterior mean

$$m_f(x) = \mathbb{E}\left[f(x) \mid \{y_i\}_{i=1}^n\right] \tag{1.13}$$

yields a smooth interpolating surface, where the height at each location is a kernel-weighted combination of nearby observations.



Thus, the RBF kernel spreads the influence of each sample point smoothly across the region, producing a continuous terrain map consistent with the measured elevations.

In many applications, the mean function is set to be identically zero, $m(x) \equiv 0$, which corresponds to placing an uninformative prior on f. Under the assumption that the mean function is identically zero, $m(x) \equiv 0$, and given a kernel $k(x_i, x_j)$, the vector of training labels satisfies

$$y \sim \mathcal{N}(0, K), \tag{1.14}$$

where K is the Gram matrix defined previously, and the mean 0 denotes the zero vector in $\mathbb{R}^n$ with the same dimension as

y.

Let $\{x_i\}_{i=1}^n$ be the training inputs and $\{y_i\}$ their corresponding observations. The above Gaussian model describes the joint distribution induced by the Gaussian process prior on the training set. For a new test point $(x_\star, y_\star)$, we can augment the training vector to obtain the joint prior

$$\begin{pmatrix} y_\star \\ y \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} k(x_\star, x_\star) & k(x_\star, \cdot)^\top \\ k(x_\star, \cdot) & K \end{pmatrix}\right), \tag{1.15}$$

where

$$k(x_\star, \cdot) = \begin{pmatrix} k(x_\star, x_1) \\ \vdots \\ k(x_\star, x_n) \end{pmatrix}. \tag{1.16}$$

To make predictions, we require the conditional distribution $p(y_\star \mid y)$. Since the joint distribution is Gaussian and the marginal $p(y)$ is known, the conditional distribution can be obtained directly using the standard formula for conditioning a multivariate normal distribution.

Jumping directly to the conclusion, let

$$x = \begin{pmatrix} x_a \\ x_b \end{pmatrix}, \qquad x_a \in \mathbb{R}^a,\ x_b \in \mathbb{R}^b, \tag{1.17}$$

and assume

$$x \sim \mathcal{N}\left(\begin{pmatrix} \mu_a \\ \mu_b \end{pmatrix}, \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix}\right). \tag{1.18}$$

Then the following standard results hold:

- **Marginalization:**

$$x_a \sim \mathcal{N}(\mu_a, \Sigma_{aa}), \qquad x_b \sim \mathcal{N}(\mu_b, \Sigma_{bb}). \tag{1.19}$$

- **Conditioning:**

$$p(x_a \mid x_b) = \mathcal{N}\big(\mu_{a|b},\ \Sigma_{a|b}\big), \tag{1.20}$$

  where the conditional mean and covariance are

$$\begin{aligned} \mu_{a|b} &= \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(x_b - \mu_b), \\ \Sigma_{a|b} &= \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}. \end{aligned} \tag{1.21}$$

Remark

Intuitively, the matrix $\Sigma_{ab}\Sigma_{bb}^{-1}$ acts as a linear projection that describes how variations in x_b influence our estimate of x_a. It adjusts the mean of x_a in proportion to how far x_b lies from its own mean.

Returning to the main track, the predictive distribution for a test label $y_\star$ conditioned on the observed vector y is

$$p(y_\star \mid y) = \mathcal{N}(y_\star \mid \mu_\star,\ \Sigma_\star)\,, \tag{1.22}$$

where the posterior mean is given by

$$\mu_\star = k(x_\star, \cdot)^\top K^{-1} y, \tag{1.23}$$

and the posterior variance is

$$\Sigma_\star = k(x_\star, x_\star) - k(x_\star, \cdot)^\top K^{-1} k(x_\star, \cdot). \tag{1.24}$$

Assume that the underlying function values y are noise-free, and that in practice we only observe

$$\hat{y} = y + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I). \tag{1.25}$$

Under this observation model, one can show that the joint distribution of the latent value $y_\star$ at a test point $x_\star$ and the

observed vector $\hat{y}$ at the training inputs is given by

Theorem 1.2

$$\begin{pmatrix} y_\star \\ \hat{y} \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} k(x_\star, x_\star) & k(x_\star, \cdot)^\top \\ k(x_\star, \cdot) & K + \sigma^2 I \end{pmatrix}\right), \tag{1.26}$$

where K denotes the Gram matrix with entries $K_{ij} = k(x_i, x_j)$.

Proof

Notice that ε is independent of $(y_\star, y)$ and $\mathbb{E}[\varepsilon] = \mathbb{E}[y_\star] = \mathbb{E}[y] = 0$:

$$\begin{aligned}
\mathrm{Cov}(y_\star, y_\star) &= \mathrm{Var}(y_\star) = k(x_\star, x_\star), \\
\mathrm{Cov}(y_\star, \hat{y}) &= \mathrm{Cov}\big(y_\star,\ y + \varepsilon\big) \\
&= \mathrm{Cov}(y_\star, y) + \mathrm{Cov}(y_\star, \varepsilon) \\
&= k(x_\star, \cdot) + 0 = k(x_\star, \cdot), \\
\mathrm{Cov}(\hat{y}, \hat{y}) &= \mathrm{Cov}\big(y + \varepsilon,\ y + \varepsilon\big) \\
&= \mathrm{Cov}(y, y) + \mathrm{Cov}(y, \varepsilon) + \mathrm{Cov}(\varepsilon, y) + \mathrm{Cov}(\varepsilon, \varepsilon) \\
&= K + 0 + 0 + \sigma^2 I_n \\
&= K + \sigma^2 I_n.
\end{aligned}$$

■

Corollary 1.1

$$y_\star \mid \hat{y} \sim \mathcal{N}(\mu_\star, \Sigma_\star) \tag{1.27}$$

where $\mu_\star = k(x_\star)^\top (K + \sigma^2 I)^{-1} \hat{y}$.

Comparing the above predictive formulas with (**??**), we arrive at a striking conclusion:

Proposition 1.1

Linear regression, when viewed from the Bayesian perspective, induces a Gaussian process.

Proof

Take any finite collection of inputs $\{x_i\}_{i=1}^n \subset \mathcal{X}$ and define

$$f_i := f(x_i) = w^\top \phi(x_i), \qquad i = 1, \ldots, n. \tag{1.28}$$

Stack these values into

$$f := \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} \in \mathbb{R}^n. \tag{1.29}$$

Let $\Phi \in \mathbb{R}^{n \times p}$ be the feature matrix

$$\Phi := \begin{pmatrix} \phi(x_1)^\top \\ \vdots \\ \phi(x_n)^\top \end{pmatrix}, \tag{1.30}$$

so that

$$f = \Phi w. \tag{1.31}$$

With the prior $w \sim \mathcal{N}(0, \sigma_w^2 I_p)$, f *is a linear transformation of a Gaussian vector, hence*

$$f \sim \mathcal{N}\big(0,\ \sigma_w^2 \Phi\Phi^\top\big). \tag{1.32}$$

Therefore, for all i, j,

$$\mathrm{Cov}\big(f(x_i), f(x_j)\big) = \big(\sigma_w^2 \Phi\Phi^\top\big)_{ij} = \sigma_w^2\, \phi(x_i)^\top \phi(x_j). \tag{1.33}$$

Define the kernel

$$k(x, x') := \sigma_w^2\, \phi(x)^\top \phi(x'). \tag{1.34}$$

Then for any finite set $\{x_i\}_{i=1}^n$, *the vector* $(f(x_1), \ldots, f(x_n))$ *is multivariate Gaussian with mean zero and covariance matrix* $[k(x_i, x_j)]_{i,j}$. *By the definition of a Gaussian process, this implies*

$$f \sim \mathcal{GP}\big(0,\ k(\cdot,\cdot)\big). \tag{1.35}$$

Now add Gaussian observation noise:

$$y_i = f(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{1.36}$$

with $\{\varepsilon_i\}$ *i.i.d. and independent of* w. *Let* $y = f + \varepsilon$ *where* $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$. *Then* y *is also Gaussian with*

$$\mathrm{Cov}(y_i, y_j) = \mathrm{Cov}\big(f(x_i), f(x_j)\big) + \mathrm{Cov}(\varepsilon_i, \varepsilon_j) \tag{1.37}$$

$$= \sigma_w^2\, \phi(x_i)^\top \phi(x_j) + \sigma^2 \delta_{ij}. \tag{1.38}$$

Hence the effective covariance function for the observed process is

$$k_{\mathrm{obs}}(x, x') = \sigma_w^2\, \phi(x)^\top \phi(x') + \sigma^2 \delta_{x,x'}, \tag{1.39}$$

which is exactly the GP regression model with kernel as stated. ■

Remark

The GP framework is powerful because it allows us to work directly with the covariance kernel $k(x, x')$ *rather than introducing a weight vector* w. *This enables the kernel trick and makes infinite-dimensional feature spaces accessible without ever defining* $\phi(x)$ *explicitly.*

Remark

In practice, Gaussian process regression requires inverting the $n \times n$ *covariance matrix* $K + \sigma^2 I$, *which has a cubic time complexity* $O(n^3)$ *and becomes prohibitive when the dataset is large. To address this issue, one typically resorts to approximation techniques such as selecting a small set of inducing points or using low-rank and sparse approximations of the kernel matrix. These methods preserve most of the predictive power of the GP while significantly reducing computational cost.*

1.2 Gaussian Process Regression (GPR)

Review the previous conclusion. Given training inputs $X = \{x_i\}_{i=1}^n$ with observations

$$y = f(X) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I), \tag{1.40}$$

and a test input $x_\star$, the joint distribution between y and the latent value $f_\star = f(x_\star)$ is

$$\begin{pmatrix} f_\star \\ y \end{pmatrix} \sim \mathcal{N}\left(0,\ \begin{pmatrix} k(x_\star, x_\star) & k(x_\star, X)^\top \\ k(x_\star, X) & K + \sigma^2 I \end{pmatrix}\right), \tag{1.41}$$

where

$$K_{ij} = k(x_i, x_j), \qquad k(x_\star, X) = \begin{pmatrix} k(x_\star, x_1) \\ \vdots \\ k(x_\star, x_n) \end{pmatrix}. \tag{1.42}$$

Conditioning on the observed vector y, the predictive distribution of $f_\star$ is Gaussian:

$$f_\star \mid y \sim \mathcal{N}(\mu_\star,\ \Sigma_\star), \tag{1.43}$$

with mean

$$\mu_\star = k(x_\star, X)^\top (K + \sigma^2 I)^{-1} y, \tag{1.44}$$

and predictive variance

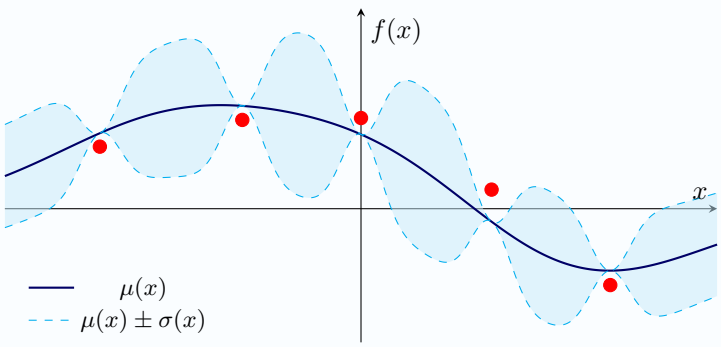
$$\Sigma_\star = k(x_\star, x_\star) - k(x_\star, X)^\top (K + \sigma^2 I)^{-1} k(x_\star, X). \tag{1.45}$$

Remark

Consider the RBF kernel

$$k_\ell(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right), \tag{1.46}$$

where $\ell > 0$ is the length-scale parameter.



For any fixed x, x', we have

$$\lim_{\ell \to 0} k_\ell(x, x') = \begin{cases} \sigma_f^2, & x = x', \\ 0, & x \neq x'. \end{cases} \tag{1.47}$$

In particular, for distinct training inputs $\{x_i\}_{i=1}^n$, the Gram matrix K_ℓ converges to a diagonal matrix:

$$\lim_{\ell \to 0} K_\ell = \sigma_f^2 I_n. \tag{1.48}$$

In the noise-free case (i.e. $\sigma^2 = 0$ in the observation model), the predictive mean at a training point x_j is

$$\mu_\ell(x_j) = k_\ell(x_j, X)^\top K_\ell^{-1} y. \tag{1.49}$$

In the limit $\ell \to 0$, we have

$$k_\ell(x_j, X) \longrightarrow \sigma_f^2 e_j, \qquad K_\ell^{-1} \longrightarrow \frac{1}{\sigma_f^2} I_n, \tag{1.50}$$

where e_j is the j-th standard basis vector in $\mathbb{R}^n$. Hence

$$\lim_{\ell \to 0} \mu_\ell(x_j) = \sigma_f^2 e_j^\top \left(\tfrac{1}{\sigma_f^2} I_n\right) y = e_j^\top y = y_j. \tag{1.51}$$

That is, in the limit $\ell \to 0$, the GP interpolates the data exactly at every training input: each training point is matched perfectly by the predictive mean.